

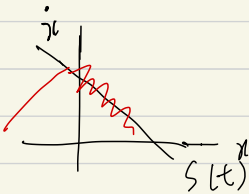
Ch 7. Sliding Control

§ 7.2 Cont. Approx. of switching control laws.

ODE + control "u"

We want to make $x(t) = x_d(t)$

Review + what we will going to do.



$$s = \dot{x} + \lambda \tilde{x}$$

* How to remove chattering

* Trade off between ϵ & Γ

$$\text{s.t. } |x - x_d| < \epsilon$$

Def.

$$\tilde{x}(t) = x(t) - x_d(t) = [\tilde{x}, \dots, \tilde{x}^{(n-1)}] : \text{Tracking error}$$

$$s(x, t) = \left(\frac{d}{dt} + \lambda \right)^{n-1} \tilde{x}$$

$$S(t) := \{ x \in \mathbb{R}^n \mid s(x, t) = 0 \}$$

$$\frac{1}{2} \frac{d}{dt} s^2 \leq -\eta |s| \quad \Leftrightarrow \quad \dot{s} \leq -\eta \operatorname{sgn}(s)$$

: sliding condition

$$\Downarrow$$
$$\text{treach} \leq \frac{|s(t=0)|}{\eta}$$

$$\begin{cases} \ddot{x} = f + u, & (n=2 \text{ case}) \\ |f - \hat{f}| \leq F & S(x, t) = \dot{\tilde{x}} + \lambda \tilde{x} \end{cases}$$

$$\dot{S} = \ddot{\tilde{x}} + \lambda \dot{\tilde{x}}$$

$$= f + u - \ddot{x}_d + \lambda \dot{\tilde{x}}$$

$$u_{eq} = -f + \ddot{x}_d - \lambda \dot{\tilde{x}} \quad (\text{we don't know } f)$$

$$\hat{u} = -\hat{f} + \ddot{x}_d - \lambda \dot{\tilde{x}}$$

$$\begin{aligned} u &:= \hat{u} - k \operatorname{sgn}(S) \\ &= -\hat{f} + \ddot{x}_d - \lambda \dot{\tilde{x}} - k \operatorname{sgn}(S) \end{aligned}$$

$$\Rightarrow \frac{1}{2} \frac{d}{dt} S^2 = S \dot{S} = S (f - \hat{f} - k \operatorname{sgn}(S))$$

$$\leq -\eta |S| \quad (\text{we want to make})$$

$$\text{Choose } k := F + \eta$$

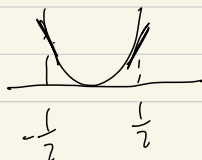
model imprecision, disturbance $\Rightarrow$ Chattering

$\operatorname{sgn}(s)$: dis continuous

Notation

$$\min f(x)$$

$$f(x) = x^2$$



$$x_{n+1} = x_n - k \sigma_k f(x_n)$$

$$k=1$$

$$x_1 = \frac{1}{2}$$

$$x_2 = -\frac{1}{2} \quad x_3 = \frac{1}{2} \dots$$

$$\ddot{x} = f + bu.$$

$$\overline{\Phi}$$

uncertainty

①

constant

f

②

$\overline{\Phi}(t)$

f

③

$\overline{\Phi}(t)$

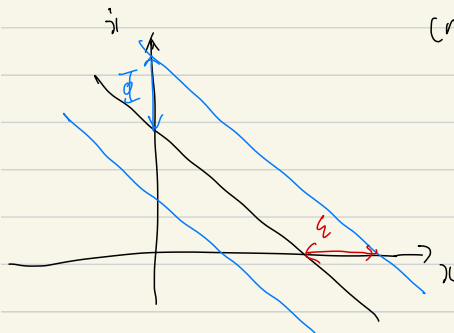
$f, b.$

1. How to eliminate chattering.

2. Trade off between precision & bandwidth.

Def. Boundary layer

$$B(t) = \{x \in \mathbb{R}^n \mid |s(x, t)| \leq \overline{\Phi}\}, \quad \overline{\Phi} > 0$$



$(n=2)$

$$\dot{\tilde{x}} + \lambda \tilde{x} = s(x, t)$$

$$\varepsilon = \frac{\overline{\Phi}}{\lambda}$$

$$\varepsilon = \frac{\overline{\Phi}}{\lambda^{n-1}}$$

(Review)

$$|s(t)| \leq \overline{\Phi} \Rightarrow \|\tilde{x}^{(n)}(t)\| \leq (2\lambda)^n \frac{\overline{\Phi}}{\lambda^{n-1}} = (2\lambda)^n \varepsilon.$$

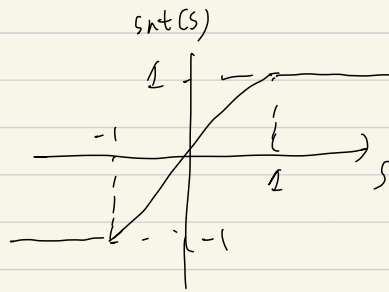
$$(\because) \left(\frac{d}{dt} + \lambda \right) y_1 = s \quad \Rightarrow \quad |y_1| \leq \frac{\Phi}{\lambda}$$

① Remove Chattering

(same example)

$$\dot{s} = f + u - \ddot{x}_d + \ddot{x}_c$$

$$u_1 := \hat{u} - k \operatorname{sat} \left(\frac{s}{\Phi} \right)$$



(inside band $\Rightarrow$ decrease the size)

$$\Rightarrow \frac{1}{2} \frac{d}{ds} s^2 = s \left(f - \hat{f} - k \operatorname{sat} \left(\frac{s}{\Phi} \right) \right)$$

$$\leq \begin{cases} -\eta |s| & s \geq \Phi \\ s \left(F - \left(F + \eta \right) \frac{s}{\Phi} \right) & s < \Phi \end{cases}$$

$$(k := F + \eta)$$

($\ddot{x}_c \geq$ Fig 7.7, Fig 7.8 Chattering removed error increased)

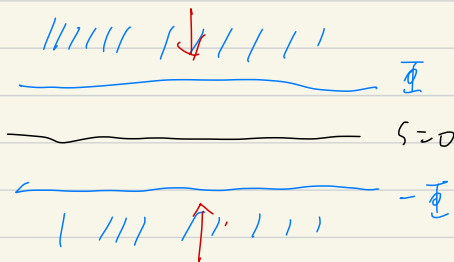
- we need to consider another sliding condition to attack inside of band.

- we want to see the relation between Φ & Σ

② $\Phi(t)$: time dependent.

$$s \geq \Phi \Rightarrow \frac{d}{dt}(s - \Phi) \leq -\eta$$

$$s \leq -\Phi \Rightarrow \frac{d}{dt}(s - (-\Phi)) \geq \eta$$



same

$$|s| \geq \Phi \Rightarrow \frac{1}{2} \frac{d}{dt} s^2 \leq (\dot{\Phi} - \eta) |s|$$

$$\dot{s} = f + u - \ddot{x}_d + \ddot{x}_r$$

$$\begin{aligned} u_2 &:= \hat{u} - \bar{k} \operatorname{sat}\left(\frac{s}{\Phi}\right) & \bar{k} &:= k - \dot{\Phi} \\ &= \hat{u} - (k - \dot{\Phi}) \operatorname{sat}\left(\frac{s}{\Phi}\right) \end{aligned}$$

$$\textcircled{3} \quad \ddot{x} = f + bu$$

$$0 < b_m \leq b \leq b_{rx} \quad \hat{b} := (b_m b_{rx})^{\frac{1}{2}}$$

$$\Rightarrow \quad \frac{1}{\beta} = \sqrt{\frac{b_m}{b_{rx}}} \leq \frac{\hat{b}}{b} \leq \sqrt{\frac{b_{rx}}{b_m}} := \beta \quad \text{중요한 것}$$

$$\frac{1}{\beta} \leq \frac{b}{\hat{b}} \leq \beta$$

(Review)

$$\dot{s} = \ddot{x} + \lambda \dot{x} = f + bu - \ddot{x}_d + \lambda \dot{x}$$

$$u_{eq} = b^{-1} (-f + \ddot{x}_d - \lambda \dot{x})$$

$$\begin{aligned} u &:= \hat{b}^{-1} (\hat{u} - k \operatorname{sgn}(s)) \\ &= \hat{b}^{-1} \left(-\hat{f} + \ddot{x}_d - \lambda \dot{x} - k \operatorname{sgn}(s) \right) \end{aligned}$$

$$\Rightarrow \dot{s} = \left(f - \frac{b}{\hat{b}} \hat{f} \right) + \left(1 - \frac{b}{\hat{b}} \right) (-\ddot{x}_d + \lambda \dot{x}) - \frac{b}{\hat{b}} k \operatorname{sgn}(s)$$

$$\text{Now } \Phi \rightarrow \Phi(u), \quad \operatorname{sgn}(s) \rightarrow \operatorname{sat}\left(\frac{s}{\Phi}\right)$$

We want to make.

$$\frac{1}{2} \frac{d}{dt} s^2 \leq \left(\frac{\dot{\Phi}}{\Phi} - \eta \right) |s| \quad \Phi \leq |s| \quad + \operatorname{sat},$$

$$\dot{s} \leq \left(\frac{\dot{\Phi}}{\Phi} - \eta \right) \operatorname{sat}\left(\frac{s}{\Phi}\right)$$

$$u_3 := \hat{b}^{-1} \left(-\hat{f} + \ddot{x}_d - \lambda \dot{x} - \bar{v} \operatorname{sat}\left(\frac{s}{\bar{\Phi}}\right) \right)$$

$$\bar{v}(\Phi) = \begin{cases} v(\Phi) - \frac{\dot{\Phi}}{\beta} & \text{when } \dot{\Phi} > 0 \\ v(\Phi) - \beta \dot{\Phi} & \dot{\Phi} < 0 \end{cases}$$

$$\Rightarrow \dot{s} = \begin{cases} \left(f - \frac{b}{\hat{b}} \hat{f} \right) + \left(1 - \frac{b}{\hat{b}} \right) (-\ddot{x}_d + \lambda \dot{x}) - \frac{b}{\hat{b}} \left(v - \frac{\dot{\Phi}}{\beta} \right) \operatorname{sat}\left(\frac{s}{\bar{\Phi}}\right) \\ \text{,,} \quad \quad \quad - \frac{b}{\hat{b}} \left(v - \beta \dot{\Phi} \right) \operatorname{sat}\left(\frac{s}{\bar{\Phi}}\right) \end{cases}$$

Inside the band. $|s| \in \bar{\Phi}$ (decay of s)

$$\Rightarrow \dot{s} = \text{,,} \quad \quad \quad - \frac{b}{\hat{b}} \bar{v} \frac{s}{\bar{\Phi}}$$

We define.

$$\frac{\bar{v}(\Phi_d)}{\bar{\Phi}} \left(\frac{b}{\hat{b}} \right)_{\max} = \lambda.$$

Balance law

$$\bar{v}(\Phi_d) = \frac{\lambda \bar{\Phi}}{\beta_d}$$

Dynamics of $\bar{\Phi}$:

by definition of $\bar{v}(\Phi)$

$$\dot{\bar{\Phi}} > 0 \quad \frac{\lambda \bar{\Phi}}{\beta_d} = v(\Phi_d) - \frac{\dot{\bar{\Phi}}}{\beta_d}$$

$$\dot{\bar{\Phi}} < 0, \quad \frac{\lambda \bar{\Phi}}{\beta_d} = v(\Phi_d) - \beta_d \dot{\bar{\Phi}}$$

$$\beta_d \bar{v}(x_d) = \lambda \bar{v} = \lambda^n \varepsilon \quad \left(\frac{\bar{v}}{\lambda^{n-1}} = \varepsilon \right)$$

$$(\text{band width})^n \times (\text{tracking precision})$$

$$\approx (\text{uncertainty along desired trajectory})$$

$$|\varepsilon| \leq \bar{v} \Rightarrow |x - x_d| \leq \frac{\bar{v}}{\lambda^{n-1}}$$

§ 7.3. The Modeling / Performance Trade-Offs

$$\lambda^n \varepsilon \approx \beta_d k_d$$

$$\Delta \varepsilon \approx 4 \left(\frac{\beta_d k_d}{\lambda^n} \right)$$

$$|F| \leq F \Rightarrow K = F + 2$$

* Better knowledge of uncertainty $\xrightarrow{f, b}$ better precision

* If λ large $\Rightarrow$ do not put much efforts to reduce β_d, k_d .

key: How large λ can be chosen,

- | | | |
|--------------------------|--|------------|
| i) structural resonant | $\lambda \leq \frac{2\pi}{3} \omega_R$ |) Hardware |
| ii) neglected time delay | $\lambda \leq \frac{1}{T_A}$ | |
| iii) sampling rate | $\lambda \leq \frac{1}{S} \omega_s$ | - Software |

i) motor

ii) actuator

iii) computation time

§ 7.4 Multi input system

$$\dot{x}_i^{(n_i)} = f_i(x) + \sum_{j=1}^m b_{ij}(x) u_j \quad (1 \leq i, j \leq 2)$$

Assume

$$|\hat{f}_i - f_i| \leq F_i$$

$$B = (I + \Delta) \hat{B}, \quad |\Delta_{ij}| \leq D_{ij}, \quad B, \hat{B} : \text{invertible}$$

Define

$$D_{ii} < 1.$$

$$s_i := \left(\frac{d}{dt} + \lambda_i \right) \tilde{x}_i$$

Want to make

$$\frac{1}{2} \frac{d}{dt} s_i^2 \leq -\eta_i |s_i|$$

$$\dot{s}_i = \ddot{x}_i + \lambda_i \dot{\tilde{x}}_i$$

$$= f_i(x) + \sum_{j=1}^m b_{ij}(x) u_j - \ddot{x}_{di} + \lambda_i \dot{\tilde{x}}_i$$

$$u := \hat{B}^{-1} \left(\ddot{x}_d - \hat{f} - \lambda \tilde{x} - K \operatorname{sgn}(s) \right)$$

$$\Rightarrow \dot{s}_i = f_i(x) - \ddot{x}_{di} + \lambda_i \dot{\tilde{x}}_i + (s_{ij} + \Delta_{ij}) \left(\ddot{x}_{dj} - \hat{f}_j - K \operatorname{sgn}(s_j) - \lambda_j \tilde{x}_j \right) \leq -\eta_i \operatorname{sgn}(s_i)$$

We can choose

$$(1 - D_{ii}) k_i + \sum_{j \in S} D_{ij} k_j = F_i + \sum_{j=1}^n D_{ij} |\dot{x}_{ij} - \hat{f}_j| + \eta_i - \lambda_j$$

Last Note.

$$\dot{x}_1 = x_2$$

$$\dot{x}_1 = x_2 + f$$

$$\dot{x}_2 = \ddot{u}$$

$$\dot{x}_2 = x + u$$

matched uncertainty

W's matched uncertainty

SKL : good.

In equal SKL